

Task-dependent neural effects of physical exercise: A systematic review and meta-analysis of functional magnetic resonance imaging studies

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ABSTRACT

Background: Physical exercise enhances cognition, likely through neuroplasticity. Whether exercise induces domain-specific brain changes according to cognitive task demands remains unclear.

Method: An Activation Likelihood Estimation (ALE) meta-analysis of 34 Functional Magnetic Resonance Imaging (fMRI) studies was conducted, categorizing studies by task type.

Results: Physical exercise increased precuneus activation, supporting a domain-general effect. Subgroup analyses revealed task-specific changes: executive tasks showed increased cingulate activation and decreased superior temporal and cerebellar activation, while low-control tasks showed increased precuneus, cuneus, and precentral activation with decreased fusiform and parahippocampal activation.

Conclusion: This meta-analysis demonstrates that physical exercise enhances task-related brain activation in domain-general hub regions such as the precuneus, while also exerting task-dependent effects in regions associated with motor and executive control. These findings suggest that physical exercise promotes both general and task-specific neural adaptations that may underlie its cognitive benefits.

1. Introduction

Cognitive functions, including the acquisition of knowledge, information processing, and reasoning, form the fundamental basis for daily activities such as social interaction, learning, and occupational performance [1]. Cognitive decline and associated neurodegenerative diseases significantly reduce individuals' quality of life and increase healthcare resource consumption and public health expenditure [2,3]. Consequently, physical exercise has gained recognition as a feasible and cost-effective non-pharmacological intervention to enhance cognitive function.

Although the cognitive benefits of physical exercise have been well established [1,2,4–69], the underlying neural mechanisms remain to be fully elucidated. Functional magnetic resonance imaging (fMRI) has emerged as a central tool for investigating the neural effects of physical

exercise. Specifically, fMRI enables the detection of physical exercise-induced changes in brain activation during cognitive tasks, offering critical insights into the functional reorganization associated with cognitive improvements [70]. Existing fMRI-based meta-analyses have demonstrated that physical exercise produces domain-general neural effects on cognition, with enhanced task-related activation notably observed in the precuneus, which correlates positively with behavioral performance gains [36]. However, systematic comparisons of physical exercise-induced activation pattern changes across different types of cognitive tasks remain scarce, limiting our understanding of whether these neural changes also exhibit domain-specific characteristics.

Different types of cognitive tasks engage distinct brain regions [19]. Tasks that require rapid responses and executive control predominantly recruit the frontoparietal executive control network, which plays a

Abbreviations: ACC, Anterior Cingulate Cortex; ALE, Activation Likelihood Estimation; BDNF, Brain-Derived Neurotrophic Factor; CEN, Central Executive Network; DMN, Default Mode Network; fMRI, Functional Magnetic Resonance Imaging; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PROSPERO, International Prospective Register of Systematic Reviews; MNI, Montreal Neurological Institute.

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central role in response inhibition and conflict monitoring [1–3,7–69, 71–82]. In contrast, tasks without overt responses or explicit executive demands typically engage regions associated with internally directed cognition and spontaneous thought, including the default mode network (DMN) [49,67]. The cognitive benefits of physical exercise may partly arise from improvements in motor control and executive functioning—core components inherently embedded in most forms of physical activity [2,6–59]. Motor control and executive functions also share overlapping neural substrates within the frontoparietal cortex [20]. These observations raise a critical question that previous research has not addressed [36]: whether the neural benefits of physical exercise are diminished during tasks lacking explicit motor or executive demands, or whether such tasks instead elicit distinct patterns of exercise-induced brain activation.

To address this unresolved question, the present study extended the dataset of Li et al. [36] by conducting an Activation Likelihood Estimation (ALE) meta-analysis focused specifically on studies directly relevant to task-dependent neural effects of physical exercise. Specifically, we compared brain activation patterns associated with physical exercise across two distinct cognitive task conditions: active executive tasks, which require overt motor responses and executive control, and non-motor low-control tasks, which involve minimal motor output and no explicit executive demands. By differentiating domain-specific from domain-general neural effects, this study seeks to elucidate the neural mechanisms underlying physical exercise-induced changes in task-related brain activation, thereby providing a more precise theoretical foundation for guiding future research.

2. Methods

Following best-practice recommendations for conducting fMRI meta-analyses (Müller et al., 2018; Tahmasian et al., 2019), ALE was employed to systematically analyze task-related brain activation changes associated with physical exercise interventions. The literature search and study selection procedures were performed in strict accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, with detailed compliance documented in Supplementary Table S1. Furthermore, the study protocol was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration number blinded for review).

2.1. Inclusion criteria

Studies included in the current meta-analysis were required to fulfill the following criteria: (1) The study must provide coordinates for brain activation changes attributable to physical exercise. Physical exercise is operationally defined as structured, intentional, and repetitive bodily movements aimed at enhancing or maintaining physical fitness components (Cooney et al., 2013); (2) Results must originate from whole-brain analyses examining changes in task-related neural activation. Included studies had to employ tasks categorized explicitly as either active executive or non-motor low-control paradigms. Active executive tasks refer to cognitive paradigms explicitly involving both overt motor responses and substantial executive control, exemplified by paradigms such as Go/No-Go, Stroop, and Flanker tasks. In contrast, non-motor low-control tasks involve minimal or no motor responses and lack explicit executive control requirements, commonly including paradigms focused on passive viewing, mental imagery, or internally directed cognitive processes; (3) The study design must include either within-subject (pre- versus post-exercise) or between-subject comparisons examining the neural effects of physical exercise; (4) In between-subject studies, control conditions could consist of either active control interventions (e.g., stretching, mild physical activities without explicit exercise components) or passive control conditions (e.g., no intervention, waiting-list controls); and (5) Only peer-reviewed publications

were considered eligible.

2.2. Study identification and data extraction

Literature searches were conducted in PubMed, Web of Science, PsycINFO, and Embase, with the final search completed on June 3, 2024. Detailed database-specific search strategies are presented in Supplementary Table S2, in which the search terms were primarily structured around three key components: cognitive function, physical exercise, and neuroimaging techniques. Comprehensive descriptions of the identification process have been documented previously [36]. All retrieved references were managed using EndNote 21, and duplicates were removed. Titles and abstracts were screened to exclude irrelevant studies, and the remaining articles underwent full-text evaluation for eligibility based on predefined inclusion criteria. Two reviewers (blinded for review) independently performed the screening and eligibility assessments, resolving discrepancies through discussion or consultation with a senior researcher (blinded for review).

Data extraction was independently conducted by the same two reviewers (blinded for review) using standardized forms, collecting information on participant characteristics, intervention protocols, and study designs, along with the coordinates of peak activation changes induced by physical exercise. For studies employing pre-post intervention designs, contrasts reflecting exercise-induced changes from baseline (post-exercise > pre-exercise) were extracted. When both exercise and control groups were available, between-group contrasts (exercise group > control group) were prioritized. Pre-intervention conditions were treated as baseline controls where applicable. For studies including multiple groups with different sample sizes, the sample size corresponding to each specific contrast was entered separately into the ALE analysis, following standard ALE meta-analysis procedures. When sample size for a specific contrast was not reported, it was estimated based on group-level data provided in the original paper.

2.3. ALE meta-analysis

ALE meta-analysis was performed using GingerALE software (version 3.0.2, BrainMap; www.brainmap.org) to identify brain regions consistently modulated by physical exercise during cognitive tasks. All reported activation coordinates were transformed into the standardized Montreal Neurological Institute (MNI) space during preprocessing using the Lancaster transform implemented in GingerALE. The ALE algorithm models each activation focus as a three-dimensional Gaussian distribution, adjusting its width based on individual study sample sizes to account for spatial uncertainty [16].

For studies reporting multiple tasks or exercise conditions, each eligible contrast comparing an exercise condition with a non-exercise control was treated as an independent experiment in accordance with standard ALE procedures. Eligible contrasts were those that directly reflected exercise-induced changes in task-related brain activation, such as exercise > control or post-exercise > pre-exercise. Initially, an overall analysis combining all eligible studies was conducted to identify common regions reflecting the domain-general neural effects of physical exercise. Subsequently, subgroup analyses were performed to determine whether activation patterns varied according to task type, highlighting potential domain-specific neural changes. Statistical significance was determined using a voxel-level threshold of $p < 0.001$ (uncorrected) with a minimum cluster extent of 100 mm³, consistent with recommendations from the GingerALE manual and previous studies with comparable sample sizes [64]. This approach was adopted to balance type I and type II error risks, as the relatively limited number of eligible studies may render permutation-based FWE correction overly conservative and reduce statistical power [65].

To ensure methodological rigor, the reporting quality of included studies was systematically evaluated using criteria adapted from prior research (Lavallée et al., 2023), covering participant characteristics,

study design, data acquisition and preprocessing, statistical analysis, and result interpretation (detailed criteria provided in Supplementary Table S3). Sensitivity analyses excluding lower-quality studies (≤ 15 criteria met) were also conducted to assess the robustness of the findings. In addition, we performed a leave-one-out sensitivity analysis, in which each study was sequentially excluded and the ALE results recalculated, to further evaluate the stability of the main findings.

3. Results

3.1. Study selection

The initial search identified 50,476 records. After removing 6343 duplicates, 44,133 records remained for screening. Title and abstract screening excluded 43,725 records that did not meet the inclusion criteria. Full-text reviews were then conducted for 408 articles retrieved from database searches, along with 31 additional records identified through citation tracking and 49 studies from previous reviews. Following comprehensive evaluation, 34 studies met the eligibility criteria and were included in the meta-analysis (see Fig. 1).

3.2. Study characteristics

Across the 34 studies included in this meta-analysis [5,10,11,13,22,29,32,34,37,38,40–42,46–48,50–56,61–63,66–68,74,79–81,83,84], a total of 943 participants (mean age = 41.1 years) were assessed. The interventions encompassed aerobic exercise (e.g. [5–13]), resistance training (e.g. [42,47]), multicomponent exercise (e.g. [63,81]), and mind–body exercise (e.g. [55]). On average, participants engaged in exercise sessions lasting approximately 41.3 min each, performed two sessions per week, and participated for an average duration of 21.6 weeks. A detailed summary of each study's design, participant characteristics, and exercise intervention parameters is provided in Supplementary Tables S4–S5.

3.3. ALE meta-analysis

3.3.1. Overall analysis

As shown in Table 1 and Fig. 2, the overall ALE meta-analysis identified a significant cluster of increased task-related brain activation following physical exercise compared to control conditions. This cluster (928 mm³) was primarily localized in the right precuneus (93.9 %), with additional contributions from the cuneus (6.1 %). In contrast, four significant clusters of decreased activation were observed. The largest cluster (560 mm³) showed reduced activation in the left superior temporal gyrus, with additional decreases found in the left precentral gyrus (224 mm³), right thalamus (208 mm³), and left posterior insular cortex (200 mm³).

3.3.2. Subgroup analysis

To investigate whether the neural effects of physical exercise vary by task type, we conducted separate subgroup ALE meta-analyses for studies employing active executive tasks and non-motor low-control tasks (Table 2).

As shown in Table 2 and Fig. 3, the subgroup analysis for active executive tasks identified two significant clusters of increased task-

Table 1

Brain regions with significant task-related activation differences between physical exercise and control conditions in overall analysis.

MNI coordinate	Mean z	Mean ALE	Size (mm ³)	Peak anatomical label
Overall analysis (Exercise > Controls)				
14, -64, 34	4.71	0.017	928	Right precuneus
Overall analysis (Exercise < Controls)				
-50, 10, -8	4.14	0.014	560	Left superior temporal gyrus
-58, 0, 24	3.71	0.012	224	Left precentral gyrus
8, -24, -4	3.57	0.011	208	Right thalamus
-48, -46, 0	3.81	0.013	200	Left posterior insular cortex

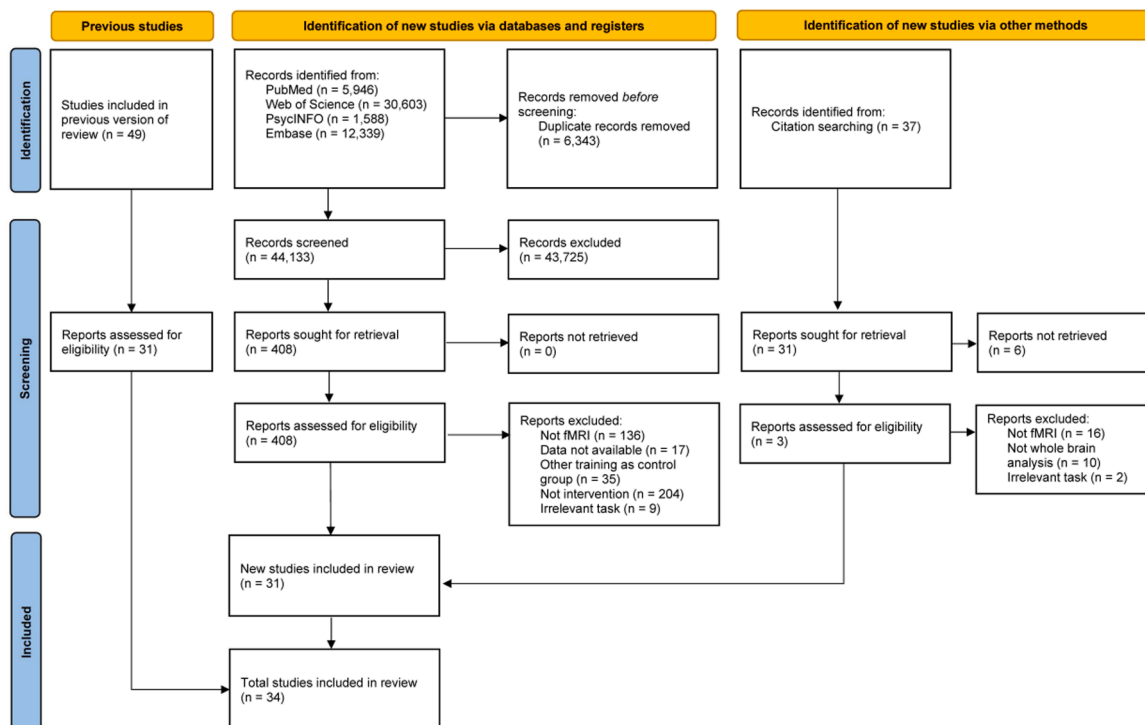


Fig. 1. PRISMA flowchart detailing the process for screening and including relevant studies.

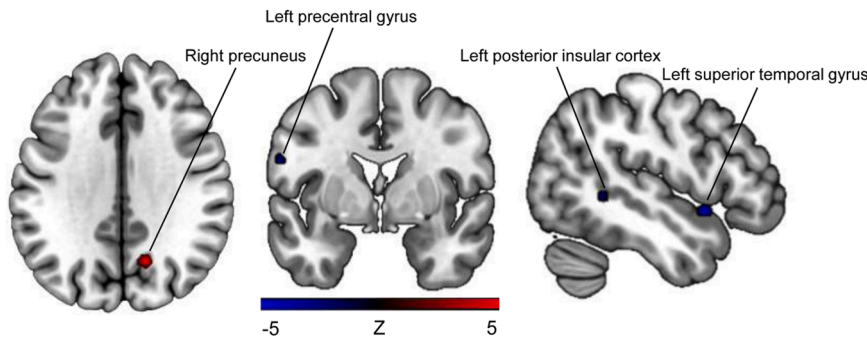


Fig. 2. Brain regions with significant task-related activation changes following physical exercise versus controls in overall analysis.

Table 2
Brain regions with significant task-related activation differences between physical exercise and control conditions in subgroup analyses.

MNI coordinate	Mean z	Mean ALE	Size (mm ³)	Peak anatomical label
Active executive tasks (Exercise > Controls)				
2, 4, 32	4.06	0.012	288	Left anterior cingulate cortex
−22, −66, 18	3.82	0.011	248	Left posterior cingulate cortex
Active executive tasks (Exercise < Controls)				
−52, 10, −8	4.21	0.012	416	Left superior temporal gyrus
8, −26, −6	4.02	0.011	352	Right red nucleus
10, −44, −22	3.61	0.009	272	Right culmen
Non-motor low-control tasks (Exercise > Controls)				
4, −66, 30	4.23	0.011	512	Right precuneus
36, −14, 62	3.81	0.009	240	Right precentral gyrus
Non-motor low-control tasks (Exercise < Controls)				
38, −52, −14	3.81	0.011	424	Right fusiform gyrus

related activation. The first cluster (288 mm³) was located in the bilateral anterior cingulate gyrus, and the second cluster (248 mm³) was located in the left posterior cingulate cortex, with partial overlap into the cuneus and precuneus. Conversely, three clusters of decreased activation were observed: the most prominent decrease was in the left superior temporal gyrus, with additional decreases in the right red nucleus (midbrain) and the right anterior cerebellar culmen.

For non-motor low-control tasks, the subgroup ALE analysis revealed two clusters of increased activation (Table 2 and Fig. 4). The first cluster (512 mm³) was located in the right precuneus, extending into the cuneus, while the second cluster (240 mm³) was located in the right precentral gyrus. In terms of decreased activation, one significant cluster (424 mm³) was identified, encompassing the right fusiform gyrus and parahippocampal gyrus.

3.3.3. Sensitivity analyses

The reporting quality of the 34 included fMRI studies was evaluated (see Supplementary Table S4). Sensitivity analyses excluding lower-quality studies (≤ 15 criteria met) showed no significant impact on the main ALE results, suggesting robustness of the findings. Consistently, the leave-one-out sensitivity analysis demonstrated that the main results remained stable after sequential exclusion of individual studies (Supplementary Table S6).

4. Discussion

The domain-general impact of physical exercise on cognition is well established. Numerous studies have reported consistent benefits of physical exercise across executive function, attention, memory, and other cognitive domains [1,45], suggesting the involvement of neural processes that extend beyond task-specific mechanisms. However, it remains unclear whether physical exercise leads to a global increase in activation across all task-relevant regions, preferentially strengthens central hub regions, or reallocates resources from task-irrelevant areas to support critical regions. Systematic investigations are required to disentangle these alternative mechanisms. Moreover, an fMRI meta-analysis has demonstrated that physical exercise consistently increases task-related activation in the precuneus, a change that is closely associated with cognitive improvements [36]. Nevertheless, whether such enhancements are accompanied by domain-specific neural adaptations remains to be determined.

Our analysis confirms that physical exercise reliably increases task-related activation in the precuneus, consistent with the meta-analytic evidence reported by Li et al. [36]. As a pivotal hub linking the DMN and the central executive network (CEN), the precuneus integrates information from multiple cortical regions and coordinates internal mentation with external task demands [17]. Long-term physical exercise increases cerebral perfusion and upregulates neurotrophic factors such as brain-derived neurotrophic factor (BDNF), thereby strengthening

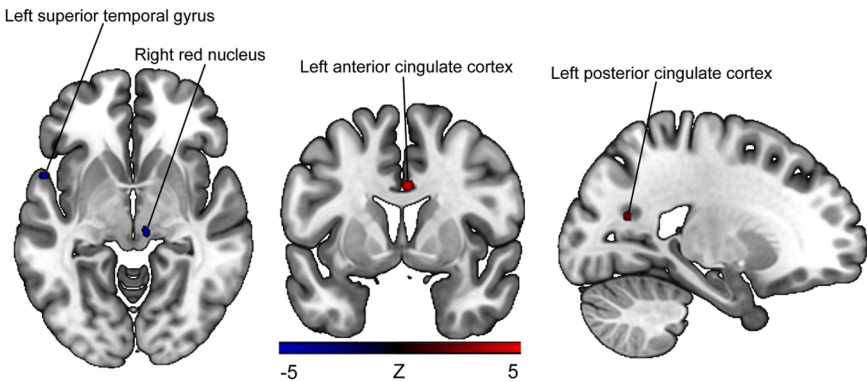


Fig. 3. Brain regions with significant task-related activation changes following physical exercise versus controls in subgroup analyses of active executive tasks.

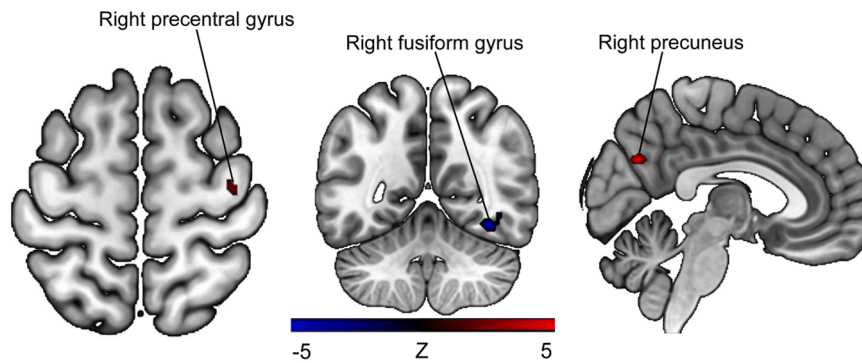


Fig. 4. Brain regions with significant task-related activation changes following physical exercise versus controls in subgroup analyses of non-motor low-control tasks.

synaptic plasticity and overall brain function [1,2,14–69,71–77]. Situated at the intersection of major networks, the precuneus is particularly sensitive to these system-wide benefits, making it a stable neural marker of exercise-induced cognitive increases [9]. These findings suggest that the precuneus may serve as a key neural substrate through which physical exercise supports domain-general cognitive improvements.

Subgroup analyses further revealed that, beyond the consistent increase in the precuneus, exercise-induced activation changes varied according to task demands for motor and executive control. Overall, increases were localized to network hubs and task-critical regions, whereas decreases were observed in areas less relevant to task performance. In active executive tasks, physical exercise increased activation in the anterior cingulate cortex and the posterior cingulate–precuneus complex, while reducing activity in auxiliary regions such as the superior temporal gyrus, red nucleus, and anterior cerebellar culmen. Increased activation of the anterior cingulate cortex has been associated with improved performance monitoring and self-regulation [75]. In contrast, in non-motor low-control tasks, physical exercise produced a “core-up, periphery-down” pattern: activation increased in the premotor cortex and precuneus but decreased in the fusiform and parahippocampal gyri [30]. This dual trajectory is consistent with the neural efficiency account, whereby reducing activity in non-essential regions and concentrating resources in hub networks increases overall processing efficiency [78]. By directly contrasting active executive and non-motor low-control tasks, our study extends previous meta-analytic work [36] by demonstrating that the neural effects of physical exercise are not only domain-general, centered on the precuneus, but also task-dependent, revealing distinct patterns of network modulation across different cognitive demands.

5. Limitations

Despite uncovering distinct patterns of neural modulation and resource reallocation across task types, this study has several limitations. First, the number of included studies was limited, making it difficult to examine how exercise protocols, participant characteristics, or analytic methods might moderate the results. Although Li et al. [36] suggested that variations in exercise parameters may not be the primary determinants of neural effects, and that participant adherence could play a more critical role, future research is still needed to systematically examine these variables in order to obtain more stable and generalizable conclusions. Second, the combination of increased and decreased activation may suggest a form of exercise-induced “resource reallocation,” but this interpretation should be treated with caution, as ALE-based findings could also be influenced by baseline variability or hemodynamic scaling. Moreover, although subgroup analyses indicated apparent differences between active executive tasks and low-control tasks, these independent tests cannot be considered a formal statistical test of interaction. Such differences may also reflect general factors such as overall engagement or arousal. Therefore, future studies should adopt

more fine-grained task designs, stricter control conditions, and analytic frameworks capable of explicitly modeling interaction effects to further test and clarify these interpretations. In addition, given the multiple exercise–control and task–control contrasts across included studies and the limited number of datasets available for each task category, the present study employed an uncorrected voxel-level threshold to balance sensitivity and specificity under conditions of limited statistical power. Future studies incorporating larger datasets should apply permutation-based or other multiple-comparison correction methods to enhance the robustness and generalizability of the findings. Finally, many of the included studies employed paradigms without overt behavioral responses and lacked follow-up assessments, thereby limiting the evaluation of performance–brain associations and the persistence of exercise-induced effects. Future meta-analyses with larger samples, richer behavioral data, longitudinal follow-up, and multimodal imaging measures are needed to address these gaps.

6. Conclusion

This meta-analysis systematically synthesized fMRI evidence to clarify how physical exercise modulates task-related brain activation across diverse cognitive paradigms. The results show that physical exercise reliably increases activation in core network hubs—particularly the precuneus—supporting a domain-general mechanism of functional gain. At the same time, physical exercise produces task-dependent regional effects: when motor control and executive demands are present, it selectively increases or decreases activity in task-relevant and auxiliary areas, suggesting a domain-specific neural impact. Together, these findings highlight both shared and distinct pathways through which physical exercise shapes brain networks to sustain and improve cognitive performance.

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Data availability

Data will be made available on request.

Ethics statement

This study is a systematic review and meta-analysis based on previously published studies and did not involve any new data collection with human participants or animals. As such, ethical approval was not required.

CRediT authorship contribution statement

Geng Li: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Yang Liu:** Writing – original draft, Visualization, Formal analysis, Data curation. **Chenzhen Liu:** Validation, Software, Investigation. **Huazhi Li:** Resources, Methodology, Investigation. **Wenhua Tang:** Validation, Formal analysis, Data curation. **Antao Chen:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.tine.2025.100274](https://doi.org/10.1016/j.tine.2025.100274).

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